HIRES: A Stiff Eight-Equation Benchmark, Solved Analytically and Verified

SIRPY Problem Library

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SIRPY · Problem 16

An ongoing series. We are building a public library of problems SIRPY has solved — each entry is one problem, the exact input we handed the solver, the exact output it returned, and an honest read of what happened. No slides. No hand-waving. Numbers you can check.

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The problem

HIRES — "High Irradiance RESponse" — is a system of eight coupled nonlinear ordinary differential equations from plant physiology, modelling how a plant's growth response is governed by light. It is one of the standard stiff test problems for initial value solvers, and it is standard precisely because it is unpleasant: the reaction rates span four orders of magnitude, so the fast components equilibrate almost immediately while the slow ones take hundreds of time units to settle.

$$
\begin{aligned}
y_1' &= -1.71\,y_1 + 0.43\,y_2 + 8.32\,y_3 + 0.0007\\
y_2' &= 1.71\,y_1 - 8.75\,y_2\\
y_3' &= -10.03\,y_3 + 0.43\,y_4 + 0.035\,y_5\\
y_4' &= 8.32\,y_2 + 1.71\,y_3 - 1.12\,y_4\\
y_5' &= -1.745\,y_5 + 0.43\,y_6 + 0.43\,y_7\\
y_6' &= -280\,y_6y_8 + 0.69\,y_4 + 1.71\,y_5 - 0.43\,y_6 + 0.69\,y_7\\
y_7' &= 280\,y_6y_8 - 1.81\,y_7\\
y_8' &= -280\,y_6y_8 + 1.81\,y_7
\end{aligned}
$$

on $0 \le t \le 321.8122$, with

$$y(0) = (1,\,0,\,0,\,0,\,0,\,0,\,0,\,0.0057)^{\mathsf{T}}.$$

The nonlinearity sits in the $280\,y_6y_8$ term, which couples the last three equations. Everything else is linear.

A conservation law, not supplied to the solver

Adding the seventh and eighth equations, the nonlinear terms and the linear terms cancel identically:

$$\frac{d}{dt}(y_7 + y_8) = \bigl(280\,y_6y_8 - 1.81\,y_7\bigr) + \bigl(-280\,y_6y_8 + 1.81\,y_7\bigr) = 0.$$

So $y_7 + y_8$ is constant for all time, equal to its initial value 0.0057. This is a property of the equations, available to anyone who looks — but it is **not** part of the input, and no solver is told it. It therefore serves as an independent audit that owes nothing to the solver's own error estimates. We use it in Section 5.

How it was entered

Eight equations, typed as written. No Jacobian, no reformulation, no stiffness flag, no tolerance.

```
sys_h <- list(
  Dy1 = "-1.71*y1 + 0.43*y2 + 8.32*y3 + 0.0007",
  Dy2 = "1.71*y1 - 8.75*y2",
  Dy3 = "-10.03*y3 + 0.43*y4 + 0.035*y5",
  Dy4 = "8.32*y2 + 1.71*y3 - 1.12*y4",
  Dy5 = "-1.745*y5 + 0.43*y6 + 0.43*y7",
  Dy6 = "-280*y6*y8 + 0.69*y4 + 1.71*y5 - 0.43*y6 + 0.69*y7",
  Dy7 = "280*y6*y8 - 1.81*y7",
  Dy8 = "-280*y6*y8 + 1.81*y7")

ic <- list(y1 = 1.0, y2 = 0.0, y3 = 0.0, y4 = 0.0,
           y5 = 0.0, y6 = 0.0, y7 = 0.0, y8 = 0.0057)

r_h <- solve_ivp(system = sys_h, x0 = 0, x1 = 321.8122,
                 ic = ic, iterations = 20, method = "sirpy")
```

`Dy1` on the left names the derivative; the string on the right is the equation exactly as printed above. That is the whole interface.

The run

```
  segment 1: [0, 0.126168]  centered at 0
  segment 2: [0.126168, 0.287297]  centered at 0.126168
  segment 3: [0.287297, 0.445883]  centered at 0.287297
  ... (3903 more segments)

INFO: the interval [0, 321.8122] exceeds a single series' reliable
      range; the solution was continued analytically over 3906
      segments (piecewise power series - still fully analytic on
      every piece).
INFO: verification by substitution - VERIFIED: every equation of the
      system is satisfied to 3.18e-09 on [0, 321.8122] (exact
      coefficient differentiation, interior points). Interface
      continuity across 3905 node(s), every component: max one-sided
      jump 5.3e-11 - measured, not assumed. Also checked: initial
      data to 0.00e+00.
solve time: 61.4 s
```

The result is not a table of numbers. It is **3906 exact Taylor polynomials**, joined end to end, each an analytic solution of the system on its own subinterval. It can be differentiated, evaluated anywhere, and — the point of this paper — substituted back into the equations symbolically.

A shorter run to $t = 10$ shows the same structure at smaller scale: 187 segments, 2.35 s, equation defect 3.17×10^{-9}, interface continuity 1.6×10^{-12} across 186 nodes.

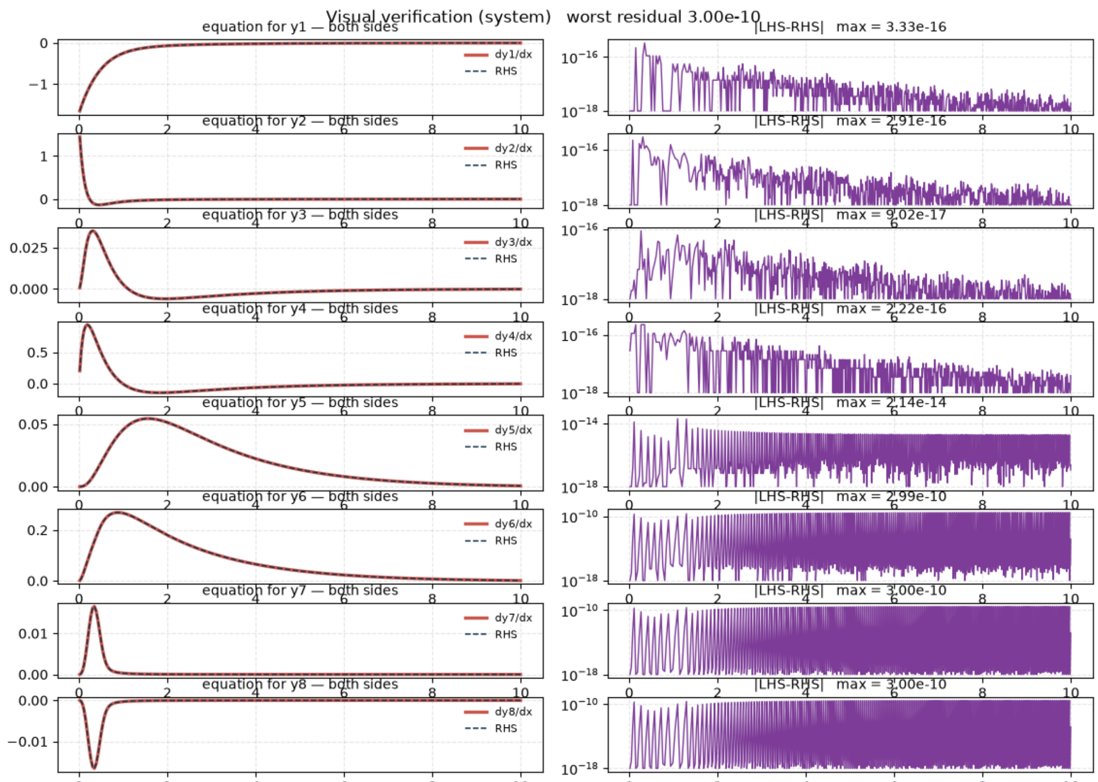

Figure 1: Verification: pointwise defect of all eight equations.

What “verified” means here, precisely

Three quantities are measured and reported, and the report also states what was **not** measured.

The equation defect, 3.18×10^{-9}. Each series is differentiated exactly — by shifting its coefficients, not by finite differences — and substituted into all eight equations. The number is the largest residual found anywhere.

Interface continuity, 5.3×10^{-11} across 3905 nodes, every component. With 3906 pieces there are 3905 interior joins, and this is where a piecewise construction can hide a defect: the equation residual is sampled at interior points, so the seams are exactly what it does not see. Each join is therefore checked separately, in all eight components, and the report says *measured, not assumed*.

The initial data, exactly.

And the limitation is stated in the same sentence as the claim: the equations are checked *at interior points, not at the interface nodes* — which is why the interface jump is a separate measurement rather than an assumption folded into the first number.

What verification does **not** establish is that this is the same trajectory another method would produce. A function can satisfy the equations and the initial data to machine precision and still have drifted onto a different solution if something has gone wrong. That requires an outside check, which is the next section.

External validation

Against an independent integrator

The same problem was integrated with SciPy’s Radau method at `rtol = 1e-13`, `atol = 1e-16` — a different algorithm, a different error control, a different implementation. At the benchmark endpoint $t = 321.8122$:

	SIRPY (analytic)	Radau (10^{-13})	difference
y_1	7.37131257380e-04	7.37131257333e-04	4.8e-14
y_2	1.44248572595e-04	1.44248572632e-04	3.7e-14
y_3	5.88872973718e-05	5.88872974097e-05	3.8e-14
y_4	1.17565134334e-03	1.17565134328e-03	5.8e-14
y_5	2.38635619909e-03	2.38635619883e-03	2.6e-13
y_6	6.23896825355e-03	6.23896825274e-03	8.1e-13
y_7	2.84999839537e-03	2.84999839519e-03	1.8e-13

y_8	2.85000160463e-03	2.85000160481e-03	1.8e-13

Every component agrees to 10^{-13} or better after 321 time units and 3905 analytic continuations. The two methods share no code path and no error estimator.

Against the conservation law

Section 1.1 showed that $y_7 + y_8$ must remain exactly 0.0057. The solver was never told this. From the returned series:

$$y_7 + y_8 = 5.7000000000000297 \times 10^{-3}$$

a drift of 2.9×10^{-17} absolute, 5.2×10^{-15} relative, accumulated across the entire interval and all 3906 pieces.

This check is the sharpest of the three, because it is the least forgiving: it depends on the two most strongly coupled components cancelling each other to fourteen digits, at every step, through the nonlinear term that makes the problem stiff in the first place. Nothing in the solver's machinery targets it.

Analytic or borrowed: a choice, stated

Called with default settings, the run above behaves differently:

```
solve time: 1.77 s
status: delegated to scipy.integrate.solve_ivp, verified
```

Rather than spend a minute building 3906 series, SIRPY hands a stiff problem to a stiff solver, **says so in the status**, and then audits the borrowed answer by substitution. Passing `method = "sirpy"` forces the analytic route — sixty-one seconds instead of two, and 3906 exact Taylor polynomials instead of an interpolated grid.

Both are legitimate; they are different products. What matters is that the output never leaves it ambiguous which one you received. The label is part of the answer.

Why this problem is a fair test

HIRES is stiff, nonlinear, and eight-dimensional, with rates spanning 0.035 to 280. Its fast transients decay within the first fraction of a time unit while the slow dynamics run for three hundred more — which is why the solver's own segmentation opens with steps of 0.126 and must sustain that resolution over the full interval, producing 3906 of them.

It is also a problem with **no closed-form solution**. Everything reported here — the defect, the interface jumps, the conservation drift, the agreement with Radau — is a measurement made without knowing the answer in advance. That is the ordinary situation for real problems, and it is the situation the verification contract is designed for.

Summary

Entered as written. Eight coupled equations typed exactly as they appear in the literature; no Jacobian, no stiffness flag, no tolerance, no reformulation.

Solved analytically. 3906 exact power-series pieces over $[0, 321.8122]$ — a differentiable object, not a table of samples.

Verified by substitution. Equation defect 3.18×10^{-9}; interface continuity 5.3×10^{-11} across 3905 joins in every component, measured rather than assumed; initial data exact. The report also states what was not checked.

Confirmed from outside, twice. Agreement with an independent Radau integration to 10^{-13} in all eight components, and preservation of a conservation law the solver was never given, to 5×10^{-15} relative.

Honest about its own choices. By default it delegates to a stiff numerical solver in under two seconds, labels the result as delegated, and audits it; on request it does the work analytically in sixty.

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Please only send problems you are free to share publicly.

SIRPY is in final validation prior to its first public release. Every line quoted above is verbatim from the solver's output; nothing is reproduced from memory.